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Adverse Weather Benchmarks and Dataset for Object Detection in Autonomous Driving

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***Abstract*—Advancements in flexible mobility solutions are directing efforts towards achieving autonomous electric driving in transport and logistics domains. Autonomous systems offer increased availability, flexibility and decreasing costs. To ensure safe and reliable behavior in autonomous driving, it is imperative to employ object detection techniques based on sensor information. Despite the significance of these techniques, limited research has been conducted to compare and assess their efficacy in adverse weather conditions like rain, fog, or snow. This paper addresses this gap by evaluating segmentation techniques on a new benchmark dataset that includes various weather scenarios. The findings underscore the impact of weather conditions on the performance of the evaluated methods, emphasizing the importance of the proposed benchmark and the analysis of solutions regarding adverse weather conditions to ensure safe and reliable behavior of autonomous vehicles.**

***Index Terms*—dataset, autonomous driving, adverse weather, object detection, semantic segmentation**

I. Introduction

Autonomous electric vehicles are on the rise in transportation and logistics, aiming to reduce individual traffic and overall congestion. These vehicles excel in routine scenarios and precise self-localization, imperative for safe and reliable operations. Nevertheless, adverse weather, communication issues, and satellite navigation limitations can disrupt perception and localization, leading to three challenges: loss of reliable navigation, false obstacle detection triggering emergency brakes, and total system dysfunction triggered by safety mechanisms. Addressing these challenges necessitates intelligent automated solutions to assess and navigate complex conditions, ensuring safety and reliability in challenging environments.

Advancing autonomous driving requires methods that perform reliably in adverse weather conditions. Accurate sensor data, free from weather-related interference, is essential to prevent false information and ensure robust performance in varied conditions. However, these methods depend on large, diverse datasets for training AI-based systems. While several datasets exist, they mainly focus on daytime driving in mild weather conditions. To address this, it's imperative to create datasets that cover diverse, challenging scenarios. Integrating multiple datasets into a unified resource is key to developing weather-agnostic technologies for autonomous vehicles. However, challenges arise from the scarcity of specific weather conditions in datasets and the variations in their design and formats. Despite the wealth of information provided by individual datasets, the disparate data structures, sensor specifications, and formats pose significant challenges for integration and analysis. Additionally, explicit weather data is limited in current datasets, which can be partially solved through

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weather databases. In [1], we outlined the creation of a combined dataset augmented with weather information for broader training and testing scenarios. This paper introduces benchmark scenarios based on common weather conditions and evaluates camera- and LiDAR-based detection models. The code for replicating the dataset will be made available on zenodo.org/communities/ci-ovgu.

II. Related Work

Multiple open datasets and challenges contribute to this field, although their coverage of various environments and conditions remains limited.

NuScenes [2] offers an autonomous driving dataset with 15 hours of data from Boston and Singapore, including LiDAR, camera, radar, and vehicle data. Although GPS data for vehicles is excluded, starting points and route layouts with GPS coordinates enable weather-related queries. The *ApolloScape* Dataset [3] contains 100,000 image frames and 80,000 LiDAR frames of urban traffic in varying conditions, accompanied by camera, LiDAR, GPS, and IMU data. The *Canadian Adverse Driving Conditions Dataset (CADCD)* [4] focuses on snowy and winter conditions, providing camera, LiDAR, GPS, and IMU data. The *Waymo Open Dataset* [5] includes labeled camera and LiDAR data, with limited IMU data collected mainly in San Francisco and Phoenix. While it lacks GPS information, it features basic weather annotations such as sunny' and rain', as well as nighttime and dawn/dusk data. The *Adverse Conditions Dataset with Correspondences for Semantic Driving Scene Understanding (ACDC)* [6] offers labeled camera images from adverse weather scenarios in Switzerland, such as rain, snow, and fog, but does not include LiDAR data. In contrast to the previously mentioned datasets and tools, [1] provides a dataset that is used as a baseline combining several datasets into a single homogenous dataset integrating additional weather information from weather databases. Currently, included are the Waymo, nuScenes (and nuImages) and CADCD datasets, with integration of further datasets and the addition of custom recorded data planned for future work.

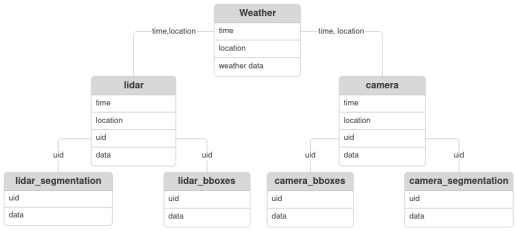

Fig. 1. Relational Schema of the Dataset

III. Combined Dataset and Benchmark Scenarios

This section details the consolidated dataset and the extracted benchmark scenarios in more detail.

A. The AdWeRSBAD Dataset

Weather details are sourced from the Open-Meteo database [7] using GPS information. Weather annotations are incorporated to enable the evaluation of methods on specific weather scenarios. The annotated data is stored in a relational database organized as shown in Fig.1, with the weather table as the primary table, identified by time and location keys. Weather conditions are recorded as strings per the WMO Code Table [8]. For samples without exact locations but with weather annotations, the data is adapted and included.

The weather table stores weather data associated with specific times and locations, facilitating the filtering of sensor data according to specific scenarios. Dedicated tables are established for each sensor type (e.g., LiDAR, camera). Although the current emphasis is on Camera and LiDAR data, future iterations may include data from any available sensors to support sensor-fusion approaches and enhance the overall versatility of the dataset. The modular design permits the usage of specific subsets of sensor data as needed. The sensor tables are linked to the weather table through time and location foreign keys, linking sensor measurements with corresponding weather conditions. This structure allows for easy retrieval of data subsets containing specific weather conditions, times of day, and sensor data. For each sensor measurement, the transformation matrix from sensor to global coordinates is stored. This facilitates the fusion of data from different sensors, such as merging all LiDAR point clouds

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into a single cloud or enhancing LiDAR data with camera pixel data and vice versa. For data consistency and convenience, all LiDAR point clouds and labels are transformed into the vehicle's coordinate system if the original dataset did not already provide the sensor data in this format. Furthermore, all provided annotation labels are redefined to provide a consistent set of labels across the unified dataset.

The retrieval of data from the database is facilitated via a PyTorch [9] iterable dataset. The dataset class also provides out-of-the-box support for common database operations and the retrieval of data for each of the different weather scenarios described in the following section.

B. Benchmark Scenarios

The scenarios described in this section can be split into 3 general groups: Scenarios focusing on the time of day, which mainly influences lighting conditions, on weather conditions, and combinations of the previous two groups. The time-of-day (ToD) scenarios consist of "Daytime", "Nighttime", and "Twilight". Each of these should provide distinct lighting conditions and is expected to mainly affect the performance of image-based approaches, although the Signal-to-noise ratio of LIDAR-based systems may also be affected [10].

For the weather scenarios, the complete set and frequency of weather patterns for samples with ground truth semantic segmentation labels in the current dataset is shown in Table I. Even though the weather information gleaned from weather databases would allow a fine-grained specification of scenarios, we use major scenarios, which will in turn be made up of groups of sub-categories, to ensure a sufficient sample size. The defined scenarios are as follows: "Sun", which includes all data without precipitation and sparse cloud cover; "Cloud", which includes all data without precipitation and dense to complete cloud cover; "Rain", which includes all samples where the major precipitation type is rain; and "Snow", which is defined analogously. The combined scenario group consists of the 12 possible combinations of the three time-of-day scenarios and the four weather scenarios.

TABLE I
SAMPLE COUNTS FOR DIFFERENT SCENARIOS WITH SEMANTIC SEGMENTATION LABELS

Weather	ToD	Camera Samples	LiDAR Samples
All	Day	208,888	151,892
All	Night	13,815	17,442
All	Twilight	13,260	13,150
Sunny	Day	158,822	135,430
Sunny	Night	13,745	13,355
Sunny	Twilight	12,880	12,650
Rainy	Day	37,234	14,151
Rainy	Night	70	2,757
Rainy	Twilight	380	500
Cloudy	Day	12,061	2,311
Cloudy	Night	0	1,330
Cloudy	Twilight	0	0
Snowy	Day	771	0
Snowy	Night	0	0
Snowy	Twilight	0	0
All	All	**235,963**	**182,484**

IV. EVALUATION

To assess the significance of the proposed dataset and benchmark, both LiDAR- and camera based semantic segmentation were evaluated on the dataset, which is currently only split into a training and a validation split, as testing splits are generally provided without ground-truth labels. For scenario-specific performance on the LiDAR data, a PointNet++ [11] semantic segmentation model is used. Due to space constraints, only a short overview of the results can be given here. The model is set up and trained according to the settings suggested by the original authors. As expected, the time of day has little impact on the LiDAR detection, especially when compared to the camera evaluation. The model achieves around 0.55 IoU for scenarios without precipitation.The notable finding is the performance during rainy scenarios, which reduce the IoU to around 0.32. This noticeable decrease in performance reinforces the hypothesis that rain, or precipitation in general, harshly affects LiDAR-based object recognition approaches. This proves that more developing effort is needed for weather-aware methods, motivating our work. For image semantic segmentation evaluation, a real-time capable ResNet-based Pyramid-Ensemble is trained [12] on

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all available data in the dataset. The models consist of a ResNet [13] and a LR-ASPP decoder [14]. Predictions from two such models – ResNet-50 and ResNet-34 – are combined for improved uncertainty estimates. Each ResNet is initialized with ImageNet pretrained weights and random weights in the decoder. The model parameters are optimized using AdamW with a learning rate of 0.001 and a weight decay of 0.02 for $50\,000$ steps using a batch size of 32. This corresponds to approximately 7 epochs.

The ensembles' averaged predictions are evaluated using mean Intersection over Union (IoU 1) to measure segmentation performance and via the Expected Calibration Error (ECE 2) following [15] to estimate the predictions (un-)certainty quality. This measure splits $[0, 1]$ into K bins. Within each bin i, the accuracy of predictions o_i is compared to the average certainty e_i. Finally, a frequency (p_i) weighted average of samples yields the ECE.

$$\text{IoU} = \frac{T_p}{T_p + F_p + F_n} \tag{1}$$

$$\text{ECE} = \sum_{i=1}^{K} p_i \cdot |o_i - e_i| \tag{2}$$

The ensemble, when trained on all available data, performs poorly on images taken exclusively at underrepresented times of day and weather conditions. This is despite the fact that all scenarios were included in the training split and therefore not due to a domain shift in the classical sense. Less common scenarios are underrepresented in the dataset, leaving the model inadequately trained to handle the different features, texture and lighting patterns appearing within them. Furthermore, the ECE is greatest where the ensemble IoU is poorest. This means that the confidence associated with the model's predictions in difficult environmental scenarios is not representative of its actual performance. As a result, it will be difficult to identify situations in which the model loses its ability to accurately detect objects based on its output activations alone.

The results indicate the performance drop per scenario can be drastic and depends on the amount of variation between training data and evaluation domain. Illumination at twilight appears to be similar, but for images captured at night, performance drops drastically.

TABLE II
CAMERA METRICS

Condition	**IoU Val, Train ↑**	**ECE Val, Train ↓**
All, Day	(0.440, 0.490)	(0.064, 0.051)
All, Night	(0.325, 0.280)	(0.116, 0.171)
All, Twilight	(0.500, 0.550)	(0.049, 0.048)
Sunny, Day	(0.503, 0.526)	(0.050, 0.049)
Sunny, Night	(0.280, 0.280)	(0.116, 0.171)
Sunny, Twilight	(0.500, 0.551)	(0.049, 0.048)
Rainy, Day	(0.272, 0.286)	(0.107, 0.094)
Rainy, Night	(——, 0.302)	(——, 0.134)
Rainy, Twilight	(——, 0.532)	(——, ——)
Cloudy, Day	(0.319, 0.321)	(0.084, 0.083)
Snowy, Day	(——, 0.196)	(——, 0.179)
Combined	(0.438, 0.481)	(0.065, 0.058)

V. CONCLUSION

This paper proposed benchmark scenarios tailored towards autonomous driving in adverse weather conditions based upon a unified dataset. By cross-referencing data from already available open datasets and enhancing them with weather annotations, this dataset substantially augments the available training data for autonomous systems operating in challenging weather scenarios. The adverse weather benchmark scenarios were defined based on typical weather scenarios, which were sufficiently present in the data. The evaluation of object recognition approaches based on LiDAR and camera data revealed a notable decline in detection quality under adverse scenarios, emphasizing the potential significance of the proposed dataset. Given the varying impact of the different scenarios on each sensor, sensor fusion approaches may be an especially promising avenue for further work. Ongoing development of the dataset and benchmarks aims to provide a more comprehensive and impactful resource for advancing research in autonomous driving under adverse conditions.

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